Pneumonia Detection from Chest X-Ray Images Using Transfer Learning with Limited Training Data

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Project Category: Healthcare, Computer Vision

This project addresses the problem of automatic pneumonia detection from chest X-ray images. The task is a binary medical image classification problem: given a chest X-ray, the model will classify it as either Normal or Pneumonia. The main focus of the project is to study a practical and important deep learning question: can data augmentation reduce the performance loss caused by limited labeled training data?

The motivation comes from the fact that deep learning models usually require large labeled datasets, while in medical imaging, collecting and labeling data is expensive and often requires expert knowledge. Therefore, understanding how well a model can perform when trained on only part of the available dataset is highly relevant for real-world healthcare applications.

The project will use a publicly available chest X-ray pneumonia dataset, such as the Kaggle Chest X-Ray Images (Pneumonia) dataset. The dataset contains labeled X-ray images from two classes, Normal and Pneumonia, and is already divided into training, validation, and test sets. No manual data collection or annotation will be required.

The proposed method is based on transfer learning using ResNet18 implemented in PyTorch. We will replace the final classification layer and fine-tune the network on the pneumonia dataset. A clear baseline will be defined by training the model without data augmentation. The proposed experimental method will train the same architecture with realistic medical-image augmentations, such as small rotations, slight translations, zoom, and possibly horizontal flip if supported by relevant literature.

The experimental design will compare three training-data sizes: 20%, 50%, and 100% of the training set. For each data size, we will run two experiments: one baseline without augmentation and one with augmentation, for a total of six main training runs. Performance will be evaluated using accuracy, precision, recall, F1-score, and confusion matrices.

Related work in medical image classification commonly uses convolutional neural networks and transfer learning for X-ray analysis. Our planned contribution is a focused empirical evaluation of how data augmentation affects pneumonia classification under limited-data conditions when using a standard transfer-learning model.

The expected outcome is an organized comparison showing how performance changes as training data decreases, whether augmentation helps recover lost performance, and under which data-size setting the improvement is most meaningful.